United States Patent	12386461
Kind Code	B2
Date of Patent	August 12, 2025
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Touch-sensing apparatus

Abstract

A touch sensing apparatus is disclosed comprising a panel having a touch surface, emitters and detectors arranged along a perimeter, a light directing portion arranged adjacent the perimeter and comprising a light directing surface, the emitters and/or the detectors are arranged opposite a rear surface of the panel to emit and/or receive light through a channel in a frame element, the channel is arranged opposite the rear surface and extends in a direction of a normal axis of the touch surface, the light directing surface and the channel are arranged on opposite sides of the panel and overlap in the direction of the plane, the light directing surface receive light from the emitters, or direct light to the detectors, through the panel and through the channel, in the direction of the normal axis. A method of manufacturing a frame element for a touch sensing apparatus is disclosed.

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Appl. No.: **18/429192**

Filed: **January 31, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240248565 A1	Jul. 25, 2024

Foreign Application Priority Data

SE	2030044-8	Feb. 10, 2020
SE	2030335-0	Nov. 09, 2020

Related U.S. Application Data

continuation parent-doc US 17759836 US 11893189 WO PCT/SE2021/050040 20210125 child-doc US 18429192

Publication Classification

Int. Cl.: **G06F3/042** (20060101)

U.S. Cl.:

CPC **G06F3/0428** (20130101); **G06F3/0421** (20130101); G06F2203/04103 (20130101)

Field of Classification Search

CPC: G06F (3/0428); G06F (3/0421); G06F (2203/04103)

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention pertains to touch-sensing apparatus that operate by propagating light above a panel. More specifically, it pertains to optical and mechanical solutions for controlling and tailoring the light paths above the panel via fully or partially randomized refraction, reflection or scattering.

Description of the Related Art

(3) In one category of touch-sensitive panels known as ‘above surface optical touch systems’, a set of optical emitters are arranged around the periphery of a touch surface to emit light that is reflected to travel and propagate above the touch surface. A set of light detectors are also arranged around the periphery of the touch surface to receive light from the set of emitters from above the touch surface. I.E., a grid of intersecting light paths are created above the touch surface, also referred to as scanlines. An object that touches the touch surface will attenuate the light on one or more scanlines of the light and cause a change in the light received by one or more of the detectors. The location (coordinates), shape or area of the object may be determined by analyzing the received light at the detectors.

(4) Optical and mechanical characteristics of the touch-sensitive apparatus affects the scattering of the light between the emitters/detectors and the touch surface, and the accordingly the detected touch signals. For example, the width of the scanlines affects touch performance factors such as detectability, accuracy, resolution, and the presence of reconstruction artefacts. Problems with previous prior art touch detection systems relate to sub-optimal performance with respect to the aforementioned factors. Further, variations in the alignment of the opto-mechanical components affects the detection process which may lead to a sub-optimal touch detection performance. Factors such as signal-to-noise ratio, detection accuracy, resolution, the presence of artefacts etc, in the touch detection process may be affected. While prior art systems aim to improve upon these factors, e.g. the detection accuracy, there is often an associated compromise in terms of having to incorporate more complex and expensive opto-mechanical modifications to the touch system. This typically results in a less compact touch system, and a more complicated manufacturing process, being more expensive.

SUMMARY OF THE INVENTION

(5) An objective is to at least partly overcome one or more of the above identified limitations of the prior art.

(6) One objective is to provide a touch-sensitive apparatus which is compact, less complex, robust and easy to assemble.

(7) Another objective is to provide an “above-surface”-based touch-sensitive apparatus with efficient use of light.

(8) One or more of these objectives, and other objectives that may appear from the description below, are at least partly achieved by means of touch-sensitive apparatuses according to the independent claims, embodiments thereof being defined by the dependent claims.

(9) According to a first aspect, a touch sensing apparatus is provided comprising a panel that defines a touch surface extending in a plane having a normal axis, a plurality of emitters and detectors arranged along a perimeter of the panel, a light directing portion arranged adjacent the perimeter and comprising a light directing surface, wherein the emitters are arranged to emit light and the light directing surface is arranged to receive the light and direct the light across the touch surface, wherein the panel comprises a rear surface, opposite the touch surface, and the emitters

and/or the detectors are arranged opposite the rear surface to emit and/or receive light through a channel in a frame element, the channel is arranged opposite the rear surface and extends in a direction of the normal axis, wherein the light directing surface and the channel are arranged on opposite sides of the panel and overlap in the direction of the plane, whereby the light directing surface receive light from the emitters, or direct light to the detectors, through the panel and through the channel, in the direction of the normal axis.
(10) According to a second aspect, a method of manufacturing a frame element for a touch sensing apparatus is provided, comprising extruding the frame element to form a light directing portion and a cavity adapted to receive a substrate comprising emitters and/or detectors, and milling a wall portion of the cavity to form a channel so that, in use, a light directing surface of the light directing portion receive light from the emitters, or direct light to the detectors, through the channel.
(11) Some examples of the disclosure provide for a more compact touch sensing apparatus.
(12) Some examples of the disclosure provide for a touch sensing apparatus that is less costly to manufacture.
(13) Some examples of the disclosure provide for a touch sensing apparatus with a reduced number of electro-optical components.
(14) Some examples of the disclosure provide for a more robust touch sensing apparatus.
(15) Some examples of the disclosure provide for a touch sensing apparatus that is more reliable to use.
(16) Some examples of the disclosure provide for reducing stray light effects.
(17) Some examples of the disclosure provide for reducing ambient light sensitivity.
(18) Some examples of the disclosure provide for a touch sensing apparatus that has a better signal-to-noise ratio of the detected light.
(19) Some examples of the disclosure provide for a touch-sensing apparatus with improved resolution and detection accuracy of small objects.
(20) Some examples of the disclosure provide for a touch-sensing apparatus with less detection artifacts.
(21) Some examples of the disclosure provide for a touch-sensing apparatus with a more uniform coverage of scanlines across the touch surface.
(22) Still other objectives, features, aspects and advantages of the present disclosure will appear from the following detailed description, from the attached claims as well as from the drawings.
(23) It should be emphasized that the term “comprises/comprising” when used in this specification is taken to specify the presence of stated features, integers, steps or components but does not preclude the presence or addition of one or more other features, integers, steps, components or groups thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) These and other aspects, features and advantages of which examples of the invention are capable of will be apparent and elucidated from the following description of examples of the present invention, reference being made to the accompanying drawings, in which;
(2) FIG. **1***a* is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;
(3) FIG. **1***b* is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;
(4) FIG. **1***c* is a schematic illustration, in a top-down view, of a touch-sensing apparatus, according to one example of the disclosure;
(5) FIGS. **1***d-e* are schematic illustrations, in top-down views, of an example of a prior art touch-

sensing apparatus;

(6) FIG. **2** is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(7) FIG. **3** is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(8) FIG. **4** is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(9) FIG. **5** is a schematic illustration, in cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(10) FIG. **6** is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(11) FIGS. **7***a-c* are schematic illustrations, in a cross-sectional side views, of a frame element for a touch-sensing apparatus, according to examples of the disclosure;

(12) FIGS. **8***a-b* are schematic illustrations, in a cross-sectional side views, of a detail of a frame element for a touch-sensing apparatus, according to examples of the disclosure;

(13) FIG. **8***c* is a schematic illustration, in a cross-sectional side views, of a detail of a frame element for a touch-sensing apparatus, according to one example of the disclosure;

(14) FIG. **8***d* are schematic illustrations of a detail of a frame element comprising a light directing surface, for a touch-sensing apparatus, in a view along the direction of a plane of a touch surface (I); a detailed section of the light directing surface (II), from the view in (I); and a side-view (III) of the section in (II), according to examples of the disclosure;

(15) FIG. **9***a* is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(16) FIG. **9***b* is a schematic illustration, in a cross-sectional side view, of a touch-sensing apparatus, according to one example of the disclosure;

(17) FIG. **10***a* is a flow-chart of a method of manufacturing a frame element for a touch sensing apparatus, according to one example of the disclosure; and

(18) FIG. **10***b* is another flow-chart of a method of manufacturing a frame element for a touch sensing apparatus, according to one example of the disclosure,

(19) FIG. **10***c* is another flow-chart of a method of manufacturing a frame element for a touch sensing apparatus, according to one example of the disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(20) In the following, embodiments of the present invention will be presented for a specific example of a touch-sensitive apparatus. Throughout the description, the same reference numerals are used to identify corresponding elements.

(21) FIG. **1***a* is a schematic illustration of a touch-sensing apparatus **100** comprising a panel **101** that defines a touch surface **102** extending in a plane **103** having a normal axis **104**. The panel **101** is a light transmissive panel. The touch-sensing apparatus **100** comprises a plurality of emitters **105** and detectors **106** arranged along a perimeter **107** of the panel **101**. FIG. **1***a* shows only an emitter **105** for clarity of presentation, while FIG. **1***b* illustrates how light is transmitted from an emitter **105** to a detector **106** across the touch surface **102**. The touch-sensing apparatus **100** comprises a light directing portion **108** arranged adjacent and along the perimeter **107**. The light directing portion **108** comprises a light directing surface **109**. The emitters **105** are arranged to emit light **110** and the light directing surface **109** is arranged to receive the light **110** and direct the light across touch surface **102** of the panel **101**. The light is reflected to detectors **106**, after propagating across the touch surface **102**, via a corresponding light directing surface **109**, as illustrated in FIG. **1***b*. FIG. **1***c* is a schematic top-down view of the touch-sensing apparatus **100**. FIG. **2** shows also schematically the reflection from the emitters **105**, and to the detectors **106**. The panel **101** comprises a rear surface **111**, opposite the touch surface **102**, and the emitters **105** and/or the detectors **106** are arranged opposite the rear surface to emit and/or receive light **110** through a

channel **112** in a frame element **113** of the touch-sensing apparatus **100**. The channel **112** is arranged opposite the rear surface **111** and extends in a direction **104′** of the normal axis **104**, i.e. essentially in parallel with the normal axis **104**. The light directing surface **109** and the channel **112** are arranged on opposite sides of the panel **101** and overlap along the direction of the plane **103**. I.e. there is an overlap in the horizontal position of the light directing surface **109** and the channel **112** in FIG. **1***a*, so that the light path may extend vertically from the light directing surface **109** to the channel **112**. The light directing surface **109** is arranged to receive light **110** from the emitters **105**, or direct light to the detectors **106**, through the panel **101** and further through the channel **112**, in the direction **104′** of the normal axis **104**. It should be understood that the main optical axis **110′** of light emission may be extending essentially along the direction **104′**, but that the light also has an angular spread around the optical axis **110′**, as indicated in FIG. **1***a*. Having the light directing surface **109** arranged above the channel **112** as exemplified in FIG. **1***a* and FIG. **4** provides for effective shielding of ambient light or system stray light. The amount of ambient or stray light reflected towards the detectors **106** may thus be minimized, and the signal to noise ratio can be improved. Having the emitters **105** emitting light **110** in the direction **104′** of the normal **104**, thus having an optical axis **110′** of the light **110** essentially parallel with the normal axis **104**, as further exemplified in FIG. **1***a* and FIG. **4**, facilitates reducing the dimensions of the touch-sensing apparatus **100** along the perimeter. The cross-sectional footprint of the emitter **105**—and detector **106** assembly may be minimized, e.g. in the direction of the plane **103** in FIGS. **1***a* and **4**. The described arrangement may also provide for minimizing angular reflections against the panel **101**, which can be advantageous in some applications. Having the light **110** propagating through the panel **101** provides for a further synergetic effect in terms of providing a compact touch-sensing apparatus **100** and minimizing the number of optical components since the panel **101** acts as a sealing element for the emitters **105** and detectors **106** from the surroundings. The panel **101** may thus act as a sealing portion to protect electronics from e.g. liquids and dust. Further optical scaling elements may thus be dispensed with. This is further advantageous in that the angle by which the light scatters across the panel **101** may be further increased, with less reflection losses, providing for an improved scanline coverage across the panel **101**. For example, Fresnel reflection losses can be minimized, as described further with respect to FIG. **1***d*. FIG. **1***d* shows an example of a prior art touch-sensing apparatus, with emitters **401** and detectors **402** arranged along sides of a touch surface **403**, and an optical sealing component **404** arranged along the sides. The optical sealing component **404** is arranged above the touch surface **403** and between the reflection surfaces on opposite sides which reflects the light across the touch surface **403** (i.e. corresponding to the position of the light directing surfaces **109**). Having such additional optical sealing component **404** may introduce unwanted reflections by the light transmitted over the touch surface **403**, in particularly when the light is reflected at high-angles along the sides of the touch surface **403**, as indicated by reflection **406** in FIG. **1***d*. FIG. **1***e* is a further detailed view of the example in FIG. **1***d*, illustrating further reflections **405**, **405′**, at each interface of such additional optical sealing component **404**, **404′**. Such reflections **405**, **405′**, could lead to significant loss of light, particularly if having the additional optical scaling component **404**, **404′**, along each side of the touch surface **403**.

(22) The reduced number of components may be particularly advantageous in some applications where additional compactness is desired. This provides also for reducing the cost of the touch-sensing apparatus **100**. As will be described in more detail below, the light directing portion **108** may be formed as part of the frame element **113** so that the light directing surface **109** is formed in the material of the frame element **113**. This further reduces the number of opto-mechanical components along the path of the light from the touch surface **102** to the emitters **105** and detectors **106**. The number of components needing alignment is thus also reduced, which simplifies assembly. A particularly compact and robust touch-sensing apparatus **100** is thus provided, with more efficient use of detection light. Touch detection performance may thus be increased, while

reducing complexity and costs.
(23) An angle (v) between the light directing surface **109** and the plane **103** of the touch surface **102** may be less than 45 degrees, as exemplified in FIG. **2** and FIG. **4**. This provides for reducing the amount of unwanted reflections of the light across the touch surface **102**, again exemplified by the reflection **406** in FIG. **1***d*, which otherwise may cause artifacts or other disturbances when detecting the attenuation of the touch signal. Unwanted light reflections may instead be reflected out of the plane **103** if having the angle (v) less than 45 degrees. The angle (v) may be in the range 41-44 degrees in some examples, for a particularly advantageous reduction unwanted light reflections. The angle (v) may be more than 45 degrees. For example, having the angle (v) in the range 46-49 degrees can also reduce unwanted reflections of the type illustrated by scanline **406** in FIG. **1***d*. It should be understood that the advantageous benefits as described above for the touch-sensing apparatus **100**, i.e. less complex, more compact and cost-effective manufacturing process, applies to examples where the angle (v) is both above 45 degrees and below 45 degrees.
(24) The panel **101** has edges **114** extending between the touch surface **102** and the rear surface **111**. The channel **112** extends in a direction parallel with the plane **102** with a width (d.sub.1) between a first channel wall **115***a*, arranged closest to a respective edge **114** of the panel **101**, and an opposite second channel wall **115***b*, as schematically illustrated in e.g. FIG. **2** and FIG. **5**. The first channel wall **115***a* may extend with an angle **116** towards the direction **104′** of the normal axis **104**. Having an angled channel wall **115***a* provides for reducing the amount of ambient light reflected towards detectors **106**. A larger portion of any ambient light being reflected against channel wall **115***a* will be reflected in a direction past the detector **106**, while also reducing light from the emitter **105** which goes directly past the light directing surface **109**, which may cause stray light problems.
(25) The light directing portion **108** has an edge portion **121** corresponding to the part of the light directing portion **108** being arranged closest to the touch surface **102**, as illustrated in FIG. **3** and FIG. **5**. The light directing surface **109** may extend from the edge portion **121** of the light directing portion **108** to a protrusion **117** of the light directing portion **108**, as schematically illustrated in FIG. **3** and FIG. **5**. The protrusion **117** may extend in a direction parallel with the plane **103** to shield ambient light from being reflected towards the channel **112**.
(26) The width (d.sub.1) of the channel **112** may be further varied for optimizing the amount of light **110** emitted towards the light directing surface **109** while providing sufficient shielding from ambient or stray light. The positions of the first and second channel walls **115***a*, **115***b*, along the direction of the plane **103** relative the emitters **105** and/or detectors **106** may be optimized depending on the particular implementation. In one example, as schematically illustrated in FIG. **3**, the position of the second channel wall **115***b* is aligned with the position of the protrusion **117** along the direction of the plane **103**. This may be particularly advantageous for shielding of ambient or stray light. At the same time blocking of the emitted light **110** may be avoided when optimizing the position of the emitters **105** relative the center of the channel **112**. The surface properties of the channel walls **115***a*, **115***b*, may also be tailored to avoid loss of detection light or reduce the impact of ambient or stray light as described further below.
(27) The emitters **105** and/or detectors **106** may be mounted to a substrate **119**. The substrate **119** may comprise a chamfered edge **120***a* to be arranged opposite a corresponding mating surface **120***b* of the frame element **113**. The mating surface **120***b* of the frame element **113** may form an angle **122** with the normal axis **104**, as schematically illustrated in FIG. **3** and FIG. **6**. The allows for effectively locking the substrate **119** in place in the correct position relative the frame element **113**. A secure alignment of the emitters **105** and/or detectors **106** relative the light directing surface **109** may thus be facilitated, which in turn facilitates signal optimization. Providing an angled surface **120***b* of the frame element **113**, as exemplified in FIG. **3**, may at the same time allow for removing at least part of the second channel wall **115**, closest to the substrate **119**, and reduce the risk of blocking light from the emitters **105**, or to the detectors **106** (see e.g. FIG. **2** versus FIG. **3**).

(28) The walls **115***a*, **115***b*, of the channel **112** may comprise a diffusive light scattering surface. The walls **115***a*, **115***b*, may thus also be utilized as reflective elements that allows better light management, e.g. recycling of light and reflecting light from lost directions towards the light directing surface **109**. A larger portion of the emitted light **110** mat thus be utilized. At the same time, the surface of the walls **115***a*, **115***b*, may be tailored to provide a specular component of the reflected light. This allows for improving the directionality of the reflected light, e.g. for directing the light towards the light directing surface **109** above the panel **101**. The ratio of the specular component of the reflected light may be varied by performing different surface treatments of the channel walls **115**, **115***b*, to affect e.g. the surface roughness thereof. The reflecting properties of the light directing surface **109** may also be varied by such surface treatments, which may comprise etching, bead blasting, sand blasting, brushing, and/or anodization, as described in more detail below.
(29) A support **123** may be attached to the substrate **119**, as exemplified in FIG. **3** and FIG. **4**. The support **123** may extending in a direction parallel with the plane **103** between the substrate **119** and a frame wall **124***a*, **124***b*, of the frame element **113**. The support **123** may facilitate alignment of the substrate **119** relative the frame element **113**. This may facilitate manufacturing and allow for accurate positioning of the emitters **105** and/or detectors **106** relative the frame element **113**.
(30) The frame element **113** may be shaped to form a cavity **125**. The emitters **105** and/or detectors **106** are mounted to a substrate **119** and the substrate **119** may be arranged in the cavity **125** so that the emitters **105** and/or detectors **106** are arranged closer to the respective edge **114** of the panel **101** than the substrate **119**, as schematically illustrated in FIG. **3**. This provides for minimizing the width of the bezel, i.e. the width of the light directing portion **108** along the direction of the plane **103**, since the emitters **105** and/or detectors **106** are placed closer to the edges **114** of the panel **101**, while maintaining the advantageous sealing effect of the panel **101** as mentioned above. A more compact touch-sensing apparatus **100** is thus provided.
(31) In particular, the cavity **125** may extend in a direction parallel with the plane **103** with a width (d.sub.2) between a first frame wall **124***a*, arranged closest to the respective edge **114** of the panel **101**, and an opposite second frame wall **124***b*. The substrate **119** may be arranged in the cavity **125** so that the emitters **105** and/or detectors **106** are arranged closer to the first frame wall **124***a* than the substrate **119**, as exemplified in FIG. **3**. This allows for minimizing the width of the bezel along the direction of the plane **103**.
(32) The substrate **119** may extend with an elongated shape in a direction **104′** of the normal axis **104**, as exemplified in FIG. **2**. This provides for reducing the dimensions of the touch sensing apparatus **100** in the direction perpendicular to the normal axis **104**, which may be desirable in some applications where the amount of space in this direction is limited, and/or when the ratio of available touch surface **102** to the surrounding frame components is to be optimized. Having the substrate **119** extending along the direction **104′** of the normal axis **104** combined with having the emitters **105** and/or detectors **106** arranged closer to the first frame wall **124***a* than the substrate **119** provides for particularly efficient utilization of space along the direction of the plane **103**.
(33) FIG. **9***a* show an example where the substrate **119** extends along the direction of the plane **103**, which provides for achieving compact dimensions along the direction of the normal axis **104**. This may be advantageous when utilized in conjunction with particularly flat display panels **301**, although the dimensions in the direction of the plane **103** around the displayed area may increase in such case. FIG. **9***b* is another example where the substrate **119** extends along the direction of the plane **103**, but the emitters **105** and/or detectors **106** are arranged to emit/receive light in the direction of the plane **103** via a reflective surface **135**. This provides for achieving compact dimensions along the direction of the normal axis **104**. The reflective surface **135** may be a specular reflective surface.
(34) The light directing surface **109** may be anodized metal. The light directing surface **109** may also be surface treated to diffusively reflect the light **110** towards the touch surface **102**. The

anodization process changes the microscopic texture of the surface **109**, and increases the thickness of the natural oxide layer on the surface **109**. The thickness and porosity of the anodized oxide surface may be varied. The anodized surface may be dyed in various colors for achieving the desired appearance. Several different colors may provide advantageous reflectance values in the infrared range, such as over 80%, for example aluminium being anodized in black, grey or silver. Other metals may also provide advantageous reflectance characteristics, such as silver. It may be particularly advantageous to use wavelengths above 940 nm where many anodized materials start to reflect significantly. Different colors may also be provided by using different alloys of e.g. aluminium. The diffusive and specular reflection components of the reflectance may be varied by performing different surface treatments of the anodized metal or alloys. The surface roughness may thus be varied to optimize the ratio of the aforementioned reflection components. The directionality of the reflected light may be increased by increasing the specular component, whereas the amount of random scattering increased with the diffusive component. For example, increasing the specular component of the reflection from the light directing surface **109** may increase the strength of the scan lines. In such case the number and/or position of the emitters **105** may be varied to compensate for any narrowing of the scanlines resulting from reduction in diffusive light scattering. Hence, in some examples, the reflective characteristics of the light directing surface **109** may be optimized, while allowing for the desired aesthetic appearance of an anodized surface.
(35) Different surface roughness characteristics may be achieved by various processes, such as etching, sand blasting, bead blasting, machining, brushing, polishing, as well as the anodization mentioned above. In one example, the light directing surface **109** may have a surface roughness defined by a slope RMS (Δq) between 0.1-0.35. The slope RMS (Δq) may be between 0.1-0.25 for an advantageous diffusivity. Higher values may decrease the strength of the signals, and too low signals may lead to a more tolerance sensitive systems where the angle (φ) by which the light is spread in the plane **103** across the touch surface **102** (as indicated by angle φ in the example of in FIG. **1***c*) is reduced and limited by emitter and detectors viewing angles. Also, scanline width may become too narrow. In another example the slope RMS (Δq) may be between 0.13-0.20 for a particularly advantageous diffusivity providing for an optimized signal strength and touch detection process, while maintaining an advantageous power consumption of the components of the touch sensing apparatus **100**.
(36) When having appropriate slope variation, the height variations of blasted or etched surface are typically in the 1 to 20 um range. However the slope RMS (Δq) optimization as described above provides for the most effective tailoring of the reflective characteristics. In some examples the light directing surface **109** has a low roughness. In one example, the light directing surface **109** may be an anodized metal surface which has not undergone any processing to increase the surface roughness. The light directing surface **109** may in such case be anodized directly after the extrusion process. The light directing surface **109** may in such case be mirror-like, i.e. the surface **109** has not undergone any processing to achieve spreading of the light. In such case the slope RMS (Δq) may be between 0-0.1, for providing a mirror-like surface. Such surface may be advantageous in applications where narrow scanlines are desired for a particular touch detection process. E.g. when it is advantageous to increase the amount of available detection light in desired directions across the touch surface **102**.
(37) The frame element **113** may comprise the light directing portion **108**. I.E., the light directing portion **108** is formed directly from the frame element **113** as an integral piece, e.g. by extrusion. The frame element **113**, and the light directing portion **108**, may be formed from various metals, such as aluminium. The light directing surface **109** may thus be an anodized metal surface of the frame element **113**. The frame element **113** may thus be utilized as a diffusive light scattering element, without having to provide a separate optical component for diffusive light scattering. The number of components may thus be reduced even further with such integrated light directing surface **109**. This further removes the need for having an additional optical sealing element to

protect such separate optical component. A more robust touch-sensing apparatus **100** which is easier to assemble is thus provided. Further, the surface of the walls **115***a*, **115***b*, of the channel **112** may be a metal surface of the frame element **113**. The reflective characteristics of the walls **115***a*, **115***b*, may be tailored as mentioned above with respect to the light directing surface **109**. The frame element **113** may form a cavity **125** in which the emitters **105** and/or detectors **106** are arranged. The frame element **113** may thus be formed as a single integral piece with light directing surfaces **109**, **115***a*, **115***b*, and cavity **125** for the substrate **119**, as well as any mounting interface **129** to a back frame **302** for a display **301**, as schematically indicated in FIG. **1***a*. This allows for minimizing the number of opto-mechanical components of the touch-sensing apparatus **100**, further providing for a particularly robust touch-sensing apparatus **100** which is less complex and more viable for mass-production.

(38) The light directing portion **108** may comprise an outer surface **126** opposite the light directing surface **109**, as indicated in e.g. FIG. **2**. The light directing surface **109** may have a higher reflectance than the outer surface **126**. Providing the light directing portion **108** of the frame element **113** with different surface treatments allows for having an effective and optimized scattering of light across the touch surface **102**, via the light directing surface **109**, while the outer surface **126** facing the user has a low reflectance for minimizing light reflections towards the user. Further, this provides for giving a desired cosmetical appearance, while not affecting the optical function (e.g. avoiding having light directing surface **109** too matte by having too high slopes. A particularly effective utilization of the manufacturing materials may thus be realized, since sections of a single integral piece of the frame element **113** may be uniquely treated to achieve the desired functionality of light reflectivity. For example, alignment of separate optical components is not needed.

(39) In one example, the walls **115***a*, **115***b* of the channel **112** may have a higher specular reflectance than the light directing surface **109**. This may provide for a more controlled reflection of emitted light towards the light directing surface **109**. The light directing surface **109** may in turn provide a larger diffusive component for broadening of the scanlines across the touch surface **102**.

(40) In one aspect a touch sensing apparatus **100** is provided, comprising a panel **101** that defines a touch surface **102** extending in a plane **103** having a normal axis **104**. A plurality of emitters **105** and detectors **106** arranged along a perimeter **107** of the panel **101**. A light directing portion **108** is arranged adjacent the perimeter **107** and comprises a light directing surface **109**. The emitters **105** are arranged to emit light **110** and the light directing surface **109** is arranged to receive the light **110** and direct the light **110** across the touch surface **102**. The panel **101** comprises a rear surface **111**, opposite the touch surface **102**. The emitters **105** and/or the detectors **106** are arranged opposite the rear surface **111** to emit and/or receive light through a channel **112** in a frame element **113**. The light directing surface **109** receive light from the emitters **105**, or direct light to the detectors **106**, through the panel **101** and through the channel **112**. The frame element **113** is formed from a metal and comprises the light directing portion **108**, where the light directing surface **109** is an anodized metal surface of the frame element **113**. The frame element **113** may also form a cavity **125** in which the emitters **105** and/or the detectors **106** are arranged so that an optical axis **110**′ of the emitted light **110** is essentially parallel with the normal axis **104**. The touch sensing apparatus **100** thus provides for the advantageous benefits as described above, by providing for a compact touch sensing apparatus **100** with improved signal to noise ratio and increased touch detection performance.

(41) FIG. **10***a* is a flowchart of a method **200** of manufacturing a frame element **113** for a touch sensing apparatus **100**. The method **200** comprises extruding **201** the frame element **113** to form a light directing portion **108** and a cavity **125** adapted to receive a substrate **119** comprising emitters **105** and/or detectors **106**. FIG. **7***a* shows an example of such extruded frame element **113**. The method **200** further comprises milling **202** a wall portion **127** of the cavity **125** to form a channel **112**. FIG. **7***a* indicates with dashed lines the wall portion **127** which is milled away, so that an open

channel **112** is provided into the cavity **125**, as exemplified in FIG. **7***b*. The channel **112** is defined by channel walls or surfaces **115***a*, **115***b*. A light directing surface **109** of the light directing portion **108** may thus receive light from the emitters **105**, or direct light to the detectors **106**, through the channel **112**, when the substrate **119** is arranged in the cavity **125**. Thus, a single integral piece of the frame element **113** may be provided, by the extrusion **201** and milling **202**, which incorporates the functions of alignment and support for the substrate **119** as well as light directing surfaces **109**, **115***a*, **115***b*. A facilitated manufacturing is provided while the structural integrity and the desired tolerances of the frame element **113** may be maintained during the process. Further, the milling **202** provides for customizing the dimensions of the channel **112**, which is difficult during the extrusion process.

(42) FIG. **7***c* shows another example of an extruded frame element **113**. The frame element **113** may be shaped so that the light directing surface **109** has a free line of sight **137**, **137′** to facilitate any subsequent surface treatment of the light directing surface **109**. The line of sight **137**, **137′**, may extend in parallel with the normal (n) of the light directing surface **109**. FIG. **7***c* shows an example where the line of sight of the light directing surface **109** is indicated with lower dashed line **137** corresponding a normal (n) to the surface **109**, and upper dashed line **137′**. Having a free line of sight **137**, **137′**, i.e. no obstruction or intersection of the described sight lines **137**, **137′**, by the frame element **113** allows for optimizing subsequent surface treatment processes of the light directing surface **109**, such as sand blasting. Obtaining the desired characteristics of the light directing surface **109** may thus be facilitated. The frame element **113** may comprise a sloped portion **138**, as schematically indicated in FIG. **7***c*, which allows for obtaining a free line of sight **137**, **137′**, of the light directing surface **109** as described above, while maintaining a compact profile of the frame element **113**. The sloped portion **138** may be arranged so that the lower sight line **137**, corresponding to the intersection point of a normal (n) with the surface **109** at a lower endpoint **139** of the surface **109** (closest to the wall portion **127**), may be extended beyond the frame element **113** without intersecting with the frame element **113** or the sloped portion **138**, as illustrated in the example of FIG. **7***c*. This provides for a facilitated access for surface treatment of the entire light directing surface **109** while maintaining a particularly compact frame element **113**. The wall portion **127** may be part of the sloped portion **138**.

(43) FIGS. **10***b-c* are further flowcharts of a method **200**. The method **200** may comprise etching, or bead- or sand blasting **2011**, **2031**, the light directing surface **109**. The light directing surface **109** may thus be provided with different reflectance characteristics. In one example the etching, or bead- or sand blasting **2011** the light directing surface **109** is performed before milling **202**, as indicated in FIG. **10***b*. The light directing surface **109** may thus be provided with different reflectance characteristics without affecting the surfaces **115***a*, **115***b*, of the channel **112**, which are shielded by the wall portion **127**, e.g. during an sand blasting process. As mentioned above, it may be advantageous to maintain a larger specular reflection component of the walls **115***a*, **115***b*, as provided after the extrusion process, while the light directing surface **109** may be subsequently treated to provide more diffusive reflection. In some examples, the etching, or bead- or sand blasting **2031** the light directing surface **109** may be performed after said milling **202**, as indicated in FIG. **10***c*. In some examples, the etching, or bead- or sand blasting **2031** the light directing surface **109** may be performed after an additional milling step **203** described below, as further indicated in FIG. **10***c*. The method **200** may comprise anodization **204** of the metal of the frame element **113**, as described above.

(44) The method **200** may comprise milling **203** a top portion **128** of the extruded light directing portion **108** so that a height (h) of the light directing portion **109** above a touch surface **102** of a panel **101**, when arranged in the frame element **113**, is reduced. FIG. **8***a* shows a detailed view of an example of the light directing portion **108** after extrusion. The radius of tip **130** of the top portion **128** is limited by the extrusion process. FIG. **8***b* show the light directing portion **108** after the top portion **128** has been milled away along the dashed line in FIG. **8***a*. The milled light

directing portion **108** has a height (h) and the corresponding tip **130′**, as indicated in FIG. **8***b*, is sharper, i.e. the radius is reduced compared to the tip **130** provided after the extrusion. Thus, a more compact light directing portion **108** has been provided, by milling away the top portion **128**, while the portion of the light directing portion **108** which is useful for reflecting the light across the touch surface **102** is essentially unaffected. The rounded tip **130** in FIG. **8***a* is not useful for directing the light across the touch surface **102**. The round tip **130** has thus been removed as illustrated in FIG. **8***b*. Milling the top portion **128** thus provides for a more effective utilization of the height of the light directing portion **108**. A sufficient height, or height distribution, of the scanline above the touch surface **102** may thus be provided, to allow for secure identification of different touch objects with different tip sizes, while minimizing the bezel height. In some examples, the height is in the range 1.5-2 mm. A height of 1.8 mm may in some examples be particularly advantageous, providing the appearance of a flush bezel.

(45) The light directing surface **109** may be concave as schematically illustrated in FIG. **8***c*. Having a light directing surface **109** which is concave towards the touch surface **102** provides for controlling the direction of the reflected light as desired and increasing the signal strength of the scanlines. The light directing surface **109** may be parabolic concave. Since the light directing surface **109** may be formed directly in the frame element **113**, as described above, the concave shape may be shaped directly in the frame element **113**. Thus, the light reflection may be controlled by shaping the frame element directly **113**, without having to introduce any additional optical component.

(46) FIG. **8***d* are schematic illustrations of a detail the light directing surface **109**, in different views I-III. The first view (I) is along the direction **103** of the plane **103**, e.g. along arrow **103** in FIG. **8***c*. The light directing surface **109** is thus illustrated as an elongated portion, arranged above the panel **101**, and the emitters **105** and detectors **106** below the panel **101**. FIG. **8***d* shows in the second view (II) a detailed section of the light directing surface **109**. The light directing surface **109** may be milled or otherwise machined to form a pattern in the surface **109**. The third view (III) of FIG. **8***d* is a cross-section along A-A in view (II), where an example of such pattern is shown, with periodic ridges **136** forming an undulating pattern or grating. Different patterns may be formed directly in the frame element **113** by milling or other machining processes, to provide a light directing surface **109** with desired reflective characteristics to control the direction of the light across the touch surface **102**.

(47) Further examples of diffusive light scattering surfaces are described in the following. Any of the diffusive light scattering surfaces described may be provided on the light directing surface **109**. The diffusive light scattering surface may be configured to exhibit at least 50% diffuse reflection, and preferably at least 70-85% diffuse reflection. Reflectivity at 940 nm above 70% may be achieved for materials with e.g. black appearance, by anodization as mentioned above (electrolytic coloring using metal salts, for example). A diffusive light scattering surface may be implemented as a coating, layer or film applied by e.g. by anodization, painting, spraying, lamination, gluing, etc. Etching and blasting as mentioned above is an effective procedure for reaching the desired diffusive reflectivity. In one example, the diffusive light scattering surface is implemented as matte white paint or ink. In order to achieve a high diffuse reflectivity, it may be preferable for the paint/ink to contain pigments with high refractive index. One such pigment is TiO.sub.2, which has a refractive index n=2.8. The diffusive light scattering surface may comprise a material of varying refractive index. It may also be desirable, e.g. to reduce Fresnel losses, for the refractive index of the paint filler and/or the paint vehicle to match the refractive index of the material on which surface it is applied. The properties of the paint may be further improved by use of EVOQUE™ Pre-Composite Polymer Technology provided by the Dow Chemical Company. There are many other coating materials for use as a diffuser that are commercially available, e.g. the fluoropolymer Spectralon, polyurethane enamel, barium-sulphate-based paints or solutions, granular PTFE, microporous polyester, GORE® Diffuse Reflector Product, Makrofol® polycarbonate films

provided by the company Bayer AG, etc. Alternatively, the diffusive light scattering surface may be implemented as a flat or sheet-like device, e.g. the above-mentioned engineered diffuser, diffuser film, or white paper which is attached by e.g. an adhesive. According to other alternatives, the diffusive light scattering surface may be implemented as a semi-randomized (non-periodic) micro-structure on an external surface possibly in combination with an overlying coating of reflective material.

(48) A micro-structure may be provided on such external surface and/or an internal surface by etching, embossing, molding, abrasive blasting, scratching, brushing etc. The diffusive light scattering surface may comprise pockets of air along such internal surface that may be formed during a molding procedure. In another alternative, the diffusive light scattering surface may be light transmissive (e.g. a light transmissive diffusing material or a light transmissive engineered diffuser) and covered with a coating of reflective material at an exterior surface. Another example of a diffusive light scattering surface is a reflective coating provided on a rough surface.

(49) The diffusive light scattering surface may comprise lenticular lenses or diffraction grating structures. Lenticular lens structures may be incorporated into a film. The diffusive light scattering surface may comprise various periodical structures, such as sinusoidal corrugations provided onto internal surfaces and/or external surfaces. The period length may be in the range of between 0.1 mm-1 mm. The periodical structure can be aligned to achieve scattering in the desired direction.

(50) Hence, as described, the diffusive light scattering surface may comprise; white- or colored paint, white- or colored paper, Spectralon, a light transmissive diffusing material covered by a reflective material, diffusive polymer or metal, an engineered diffuser, a reflective semi-random micro-structure, in-molded air pockets or film of diffusive material, different engineered films including e.g. lenticular lenses, or other micro lens structures or grating structures. The diffusive light scattering surface preferably has low NIR absorption.

(51) In a variation of any of the above embodiments wherein the diffusive light scattering element provides a reflector surface, the diffusive light scattering element may be provided with no or insignificant specular component. This may be achieved by using either a matte diffuser film in air, an internal reflective bulk diffusor or a bulk transmissive diffusor. This allows effective scanline broadening by avoiding the narrow, super-imposed specular scanline usually resulting from a diffusor interface having a specular component, and providing only a broad, diffused scanline profile. By removing the super-imposed specular scanline from the touch signal, the system can more easily use the broad, diffused scanline profile. Preferably, the diffusive light scattering surface has a specular component of less than 1%, and even more preferably, less than 0.1%. Alternatively, where the specular component is greater than 0.1%, the diffusive light scattering element is preferably configured with surface roughness to reduce glossiness, e.g. micro structured.

(52) The panel **101** may be made of glass, poly(methyl methacrylate) (PMMA) or polycarbonates (PC). The panel **101** may be designed to be overlaid on or integrated into a display device or monitor (not shown). It is conceivable that the panel **101** does not need to be light transmissive, i.e. in case the output of the touch does not need to be presented through panel **101**, via the mentioned display device, but instead displayed on another external display or communicated to any other device, processor, memory etc. The panel **101** may be provided with a shielding layer such as a print, i.e. a cover with an ink, to block unwanted ambient light. The amount of stray light and ambient light that reaches the detectors **106** may thus be reduced.

(53) As used herein, the emitters **105** may be any type of device capable of emitting radiation in a desired wavelength range, for example a diode laser, a VCSEL (vertical-cavity surface-emitting laser), an LED (light-emitting diode), an incandescent lamp, a halogen lamp, etc. The emitter **105** may also be formed by the end of an optical fiber. The emitters **105** may generate light in any wavelength range. The following examples presume that the light is generated in the infrared (IR), i.e. at wavelengths above about 750 nm. Analogously, the detectors **106** may be any device capable of converting light (in the same wavelength range) into an electrical signal, such as a photo-

detector, a CCD device, a CMOS device, etc.
(54) With respect to the discussion above, “diffuse reflection” refers to reflection of light from a surface such that an incident ray is reflected at many angles rather than at just one angle as in “specular reflection”. Thus, a diffusively reflecting element will, when illuminated, emit light by reflection over a large solid angle at each location on the element. The diffuse reflection is also known as “scattering”. The described examples refer primarily to aforementioned elements in relation to the emitters **105**, to make the presentation clear, although it should be understood that the corresponding arrangements may also apply to the detectors **106**.
(55) The invention has mainly been described above with reference to a few embodiments. However, as is readily appreciated by a person skilled in the art, other embodiments than the ones disclosed above are equally possible within the scope and spirit of the invention, which is defined and limited only by the appended patent claims.
(56) For example, the specific arrangement of emitters and detectors as illustrated and discussed in the foregoing is merely given as an example. The inventive coupling structure is useful in any touch-sensing system that operates by transmitting light, generated by a number of emitters, across a panel and detecting, at a number of detectors, a change in the received light caused by an interaction with the transmitted light at the point of touch.

Claims

1. A touch sensing apparatus comprising: a panel that defines a touch surface extending in a plane having a normal axis, a plurality of emitters and detectors arranged along a perimeter of the panel, and a light directing portion arranged adjacent the perimeter and comprising a light directing surface, wherein the emitters are arranged to emit light and the light directing surface is arranged to receive the light and direct the light across the touch surface, wherein the panel comprises a rear surface, opposite the touch surface, and the plurality of emitters that are configured to emit light and/or the plurality of detectors that are configured to receive light through a channel in a frame element formed from a metal are arranged opposite the rear surface, the channel is arranged opposite the rear surface and extends in a direction of the normal axis, wherein the light directing surface and the channel are arranged on opposite sides of the panel and overlap in the direction of the plane, and whereby the light directing surface receives light from the emitters, or directs light to the plurality of detectors, through the panel and through the channel, in the direction of the normal axis, wherein the light directing surface is a metal surface of the frame element, wherein the light directing portion comprises an outer surface opposite the light directing surface, wherein the light directing surface has a higher reflectance than the outer surface.
2. The touch sensing apparatus according to claim 1, wherein the light directing surface is an anodized metal.
3. The touch sensing apparatus according to claim 2, wherein the frame element is formed from said metal.
4. The touch sensing apparatus according to claim 1, wherein the light directing surface is an etched metal surface, sand blasted metal surface, bead blasted metal surface, or brushed metal surface of the frame element for increasing surface roughness.
5. The touch sensing apparatus according to claim 1, wherein the light directing surface diffusively reflects the light over the touch surface.
6. The touch sensing apparatus according to claim 1, the panel having edges extending between the touch surface and the rear surface, the channel extending in a direction parallel with the plane with a width (d.sub.1) between a first channel wall, arranged closest to a respective edge of the panel, and an opposite second channel wall, wherein the first channel wall extends with an angle towards the direction of the normal axis.
7. The touch sensing apparatus according to claim 1, wherein the light directing surface extends

from an edge portion of the light directing portion being arranged closest to the touch surface to a protrusion of the light directing portion, the protrusion extending in a direction parallel with the plane to shield ambient or stray light from being reflected towards the channel.
8. The touch sensing apparatus according to claim 1, wherein walls of the channel comprise a diffusive light scattering surface.
9. The touch sensing apparatus according to claim 1, wherein the emitters and/or detectors are mounted to a substrate, wherein a support is attached to the substrate, the support extending in a direction parallel with the plane between the substrate and a frame wall of the frame element.
10. The touch sensing apparatus according to claim 1, wherein the panel having edges extending between the touch surface and the rear surface, wherein the frame element forms a cavity, and wherein the plurality of emitters are mounted to a substrate and the substrate is arranged in the cavity so that the plurality of emitters are arranged closer to the respective edge of the panel than the substrate, or wherein the plurality of detectors are mounted to the substrate and the substrate is arranged in the cavity so that the plurality of emitters are arranged closer to the respective edge of the panel than the substrate.
11. The touch sensing apparatus according to claim 10, wherein the cavity extends in a direction parallel with the plane with a width (d.sub.2) between a first frame wall, arranged closest to the respective edge of the panel, and an opposite second frame wall, wherein the substrate is arranged in the cavity so that the emitters and/or detectors are arranged closer to the first frame wall than the substrate.
12. The touch sensing apparatus according to claim 10, wherein the substrate extends with an elongated shape in a direction of the normal axis.
13. The touch sensing apparatus according to claim 1, wherein the metal surface is concave towards the touch surface.
14. The touch sensing apparatus according to claim 1, wherein the light directing surface has a surface roughness defined by a slope RMS (Δq) between 0.1-0.25.
15. The touch sensing apparatus according to claim 14, wherein the light directing surface has a surface roughness defined by a slope RMS (Δq) between 0.13-0.20.
16. The touch sensing apparatus according to claim 1, wherein the light directing surface has a surface roughness defined by a slope RMS (Δq) between 0-0. 1.
17. The touch sensing apparatus according to claim 1, wherein walls of the channel have a higher specular reflectance than the light directing surface.
18. A touch sensing apparatus comprising: a panel that defines a touch surface extending in a plane having a normal axis, a plurality of emitters and detectors arranged along a perimeter of the panel, a light directing portion arranged adjacent the perimeter and comprising a light directing surface, wherein the plurality of emitters are arranged to emit light and the light directing surface is arranged to receive the light and direct the light across the touch surface, the panel comprises a rear surface, opposite the touch surface, the plurality of emitters that are configured to emit light and/or the plurality of detectors that are configured to receive light through a channel in a frame element formed from a metal are arranged opposite the rear surface, the light directing surface receives light from the emitters, or directs light to the plurality of detectors, through the panel and through the channel, wherein the frame element is formed from a metal and comprises the light directing portion, and the light directing surface is an anodized metal surface of the frame element, wherein the light directing portion comprises an outer surface opposite the light directing surface, wherein the light directing surface has a higher reflectance than the outer surface.
19. A method of manufacturing a frame element of metal for a touch sensing apparatus, comprising extruding the frame element to form a light directing portion being a metal surface of the frame element, milling a wall portion of the frame element to form a channel so that, in use, a light directing surface of the light directing portion receives light from a plurality of emitters, or directs light to a plurality of detectors, through the channel, the light directing portion comprises an outer

surface opposite the light directing surface, wherein the light directing surface has a higher reflectance than the outer surface.
20. The method according to claim 19, comprising etching, or bead-blasting, or sand blasting the light directing surface.
21. The method according to claim 20, comprising milling a top portion of the extruded light directing portion so that a height (h) of the light directing portion above a touch surface of a panel, when arranged in the frame element, is reduced.